Dongze (Zed) Li

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Education

University of Waterloo — Expected May 2027
Master of Engineering in Electrical and Computer Engineering — *GPA 85.6/100*

Hebei Normal University — Graduated June 2025
Bachelor of Engineering in Computer Science and Technology — ***Ranked 1st**, GPA 3.92/4.0*

Technical Skills

Languages: Java, Python, JavaScript/TypeScript, SQL
Frameworks & Testing: Spring Boot, React, Next.js, pandas, JUnit, pytest
Systems & Databases: Redis, RocketMQ, MySQL, PostgreSQL, SQL Server, REST APIs, ETL
Cloud & Tools: AWS, OpenShift, Docker, Linux, GitHub Actions, Jenkins, Codex CLI, Claude Code

Experience

IT Trainee (Co-op Student) — May 2026 – Present
City of Toronto — *Toronto, ON*

- Designed and implemented an end-to-end **Python/SQL processing system** for **47,904 work orders** across 200+ vendors, automatically detecting 10% with overlapping activities, billing anomalies, or inconsistent records
- Built an **idempotent validation and repair system** using state-transition rules and guarded database writes, correcting **60,000+ inconsistent status records** between internal systems and SAP
- Developed a Python-based batch ingestion workflow to submit **240,000 attachment metadata records** through REST APIs, pacing 3,000 records every 20 minutes to accommodate downstream ETL processing
- Refactored validation rules into reusable Python modules and added **pytest** coverage to improve pipeline reliability and maintainability

Java Backend Intern — Dec 2024 – May 2025
Beijing Ainfinit Technology Co., Ltd — *Shijiazhuang, China*

- Designed a **RocketMQ-based event-driven logging architecture**, decoupling device logging from synchronous command execution and reducing command latency by **80%**
- Designed and implemented an internal notification service for a smart-vending video-annotation platform, delivering reliable task and reward alerts for **2,000+ daily active users**

Projects

High-Concurrency Coupon Distribution System | *Java, Spring Boot, Redis, MySQL, Lua* — fortii2/bitelog

- Engineered a high-concurrency coupon distribution service using Redis and Lua to enforce atomic inventory updates and idempotent request processing
- Reduced **P95 request latency by 73%** (723 ms to 193 ms) by decoupling order processing through Redis Streams

Multi-Layer Self-Adaptive Web System | *OpenShift, Sysdig, Java* — fortii2/MAPEK

- Built an automated monitoring and horizontal-scaling system on OpenShift using Sysdig telemetry and application performance metrics
- Designed adaptive scaling and cooldown policies that reduced peak response time by **60%** and error rate by **44%** under 1,000-thread stress tests

Campus Utility Telegram Bot | *Java, Spring Boot, React, AWS, Docker, LLM* — fortii2/watloo

- Developed a Spring Boot application integrating Telegram and OpenAI APIs for **LLM-based correction**, improving command handling robustness
- Automated containerized deployment to AWS EC2 using GitHub Actions, Docker, and GHCR, establishing an end-to-end **CI/CD pipeline**